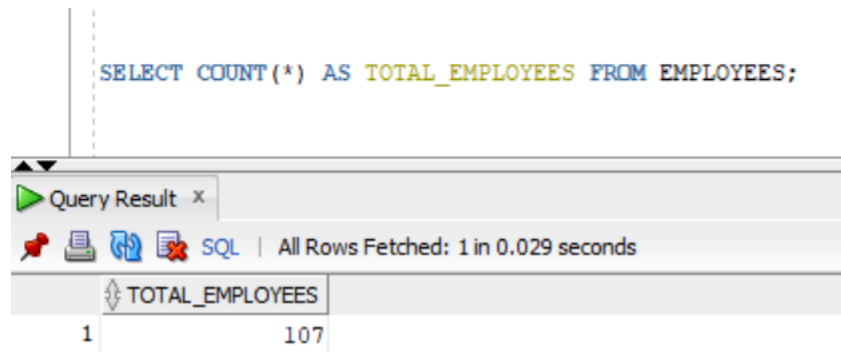


Aggregation Functions in Oracle SQL - HR Schema

1. COUNT: total number of employees

SELECT COUNT(*) AS TOTAL_EMPLOYEES FROM EMPLOYEES;

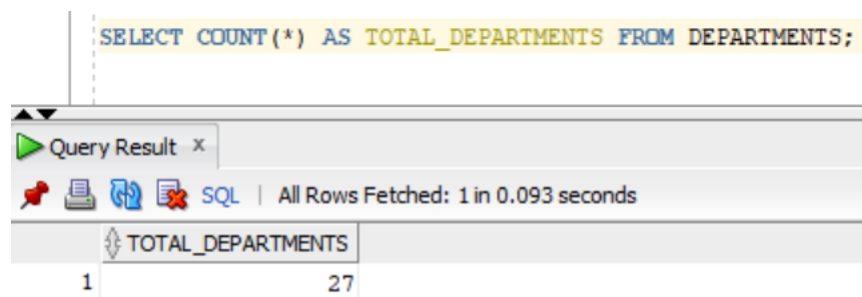


The screenshot shows a SQL Developer window with the query `SELECT COUNT(*) AS TOTAL_EMPLOYEES FROM EMPLOYEES;` entered in the SQL editor. Below the editor, the 'Query Result' tab is active, displaying the results of the query. The status bar indicates 'All Rows Fetched: 1 in 0.029 seconds'. The result set is a single row with the column header 'TOTAL_EMPLOYEES' and the value '107'.

TOTAL_EMPLOYEES
107

2. COUNT: total number of departments

SELECT COUNT(*) AS TOTAL_DEPARTMENTS FROM DEPARTMENTS;

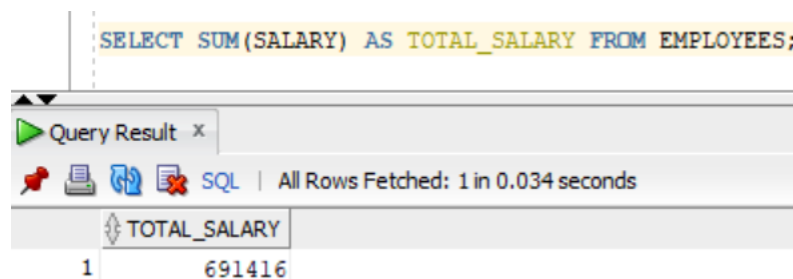


The screenshot shows a SQL Developer window with the query `SELECT COUNT(*) AS TOTAL_DEPARTMENTS FROM DEPARTMENTS;` entered in the SQL editor. Below the editor, the 'Query Result' tab is active, displaying the results of the query. The status bar indicates 'All Rows Fetched: 1 in 0.093 seconds'. The result set is a single row with the column header 'TOTAL_DEPARTMENTS' and the value '27'.

TOTAL_DEPARTMENTS
27

3. SUM: total salary of all employees

SELECT SUM(SALARY) AS TOTAL_SALARY FROM EMPLOYEES;

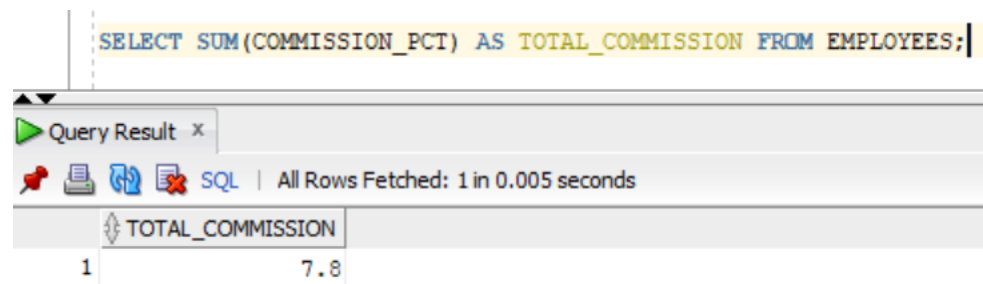


The screenshot shows a SQL Developer window with the query `SELECT SUM(SALARY) AS TOTAL_SALARY FROM EMPLOYEES;` entered in the SQL editor. Below the editor, the 'Query Result' tab is active, displaying the results of the query. The status bar indicates 'All Rows Fetched: 1 in 0.034 seconds'. The result set is a single row with the column header 'TOTAL_SALARY' and the value '691416'.

TOTAL_SALARY
691416

4. SUM: total commission percentage

SELECT SUM(COMMISSION_PCT) AS TOTAL_COMMISSION FROM EMPLOYEES;

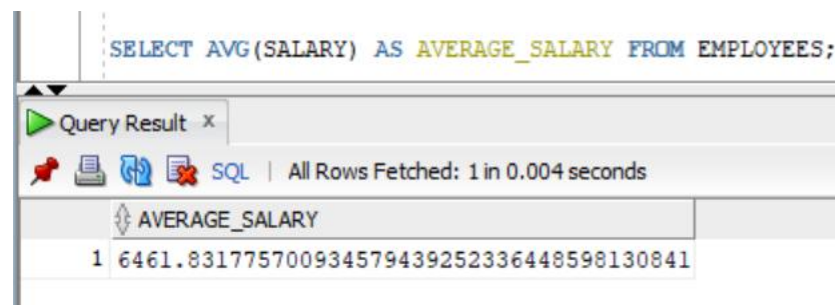


The screenshot shows a SQL query window with the following text: `SELECT SUM(COMMISSION_PCT) AS TOTAL_COMMISSION FROM EMPLOYEES;`. Below the query, a 'Query Result' window displays the results. It indicates 'All Rows Fetched: 1 in 0.005 seconds'. The result table has one column, 'TOTAL_COMMISSION', and one row with the value '7.8'.

	TOTAL_COMMISSION
1	7.8

5. AVG: average salary of all employees

SELECT AVG(SALARY) AS AVERAGE_SALARY FROM EMPLOYEES;

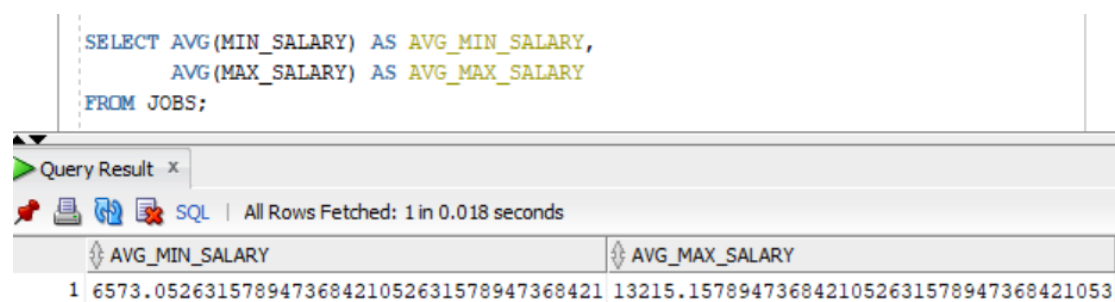


The screenshot shows a SQL query window with the following text: `SELECT AVG(SALARY) AS AVERAGE_SALARY FROM EMPLOYEES;`. Below the query, a 'Query Result' window displays the results. It indicates 'All Rows Fetched: 1 in 0.004 seconds'. The result table has one column, 'AVERAGE_SALARY', and one row with a long decimal value.

	AVERAGE_SALARY
1	6461.831775700934579439252336448598130841

6. AVG: average minimum and maximum salary from JOBS

SELECT AVG(MIN_SALARY) AS AVG_MIN_SALARY, AVG(MAX_SALARY) AS
AVG_MAX_SALARY FROM JOBS;

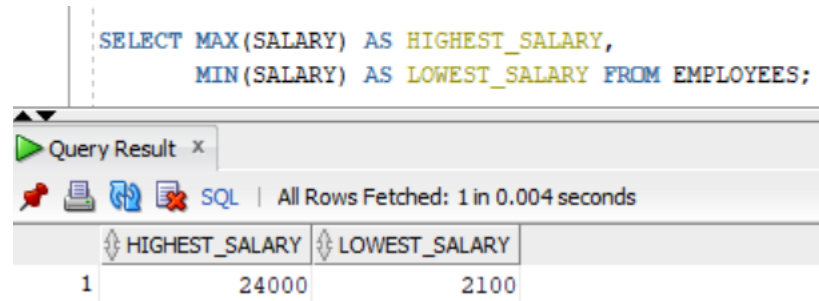


The screenshot shows a SQL query window with the following text: `SELECT AVG(MIN_SALARY) AS AVG_MIN_SALARY, AVG(MAX_SALARY) AS AVG_MAX_SALARY FROM JOBS;`. Below the query, a 'Query Result' window displays the results. It indicates 'All Rows Fetched: 1 in 0.018 seconds'. The result table has two columns, 'AVG_MIN_SALARY' and 'AVG_MAX_SALARY', and one row with two long decimal values.

	AVG_MIN_SALARY	AVG_MAX_SALARY
1	6573.052631578947368421052631578947368421	13215.1578947368421052631578947368421053

7. MAX and MIN: highest and lowest salary

```
SELECT MAX(SALARY) AS HIGHEST_SALARY, MIN(SALARY) AS LOWEST_SALARY  
FROM EMPLOYEES;
```

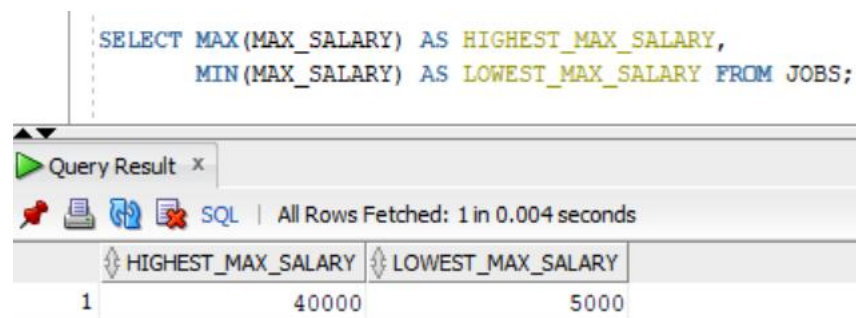


The screenshot shows a SQL query window with the following text: `SELECT MAX(SALARY) AS HIGHEST_SALARY, MIN(SALARY) AS LOWEST_SALARY FROM EMPLOYEES;`. Below the query, a 'Query Result' window displays the results. It shows a single row with two columns: 'HIGHEST_SALARY' with a value of 24000 and 'LOWEST_SALARY' with a value of 2100. The status bar indicates 'All Rows Fetched: 1 in 0.004 seconds'.

	HIGHEST_SALARY	LOWEST_SALARY
1	24000	2100

8. MAX and MIN: highest and lowest max salary from JOBS

```
SELECT MAX(MAX_SALARY) AS HIGHEST_MAX_SALARY,  
MIN(MAX_SALARY) AS LOWEST_MAX_SALARY FROM JOBS;
```

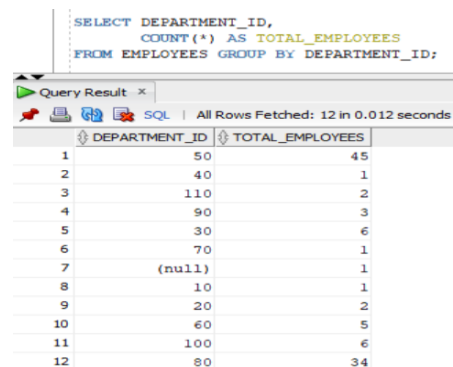


The screenshot shows a SQL query window with the following text: `SELECT MAX(MAX_SALARY) AS HIGHEST_MAX_SALARY, MIN(MAX_SALARY) AS LOWEST_MAX_SALARY FROM JOBS;`. Below the query, a 'Query Result' window displays the results. It shows a single row with two columns: 'HIGHEST_MAX_SALARY' with a value of 40000 and 'LOWEST_MAX_SALARY' with a value of 5000. The status bar indicates 'All Rows Fetched: 1 in 0.004 seconds'.

	HIGHEST_MAX_SALARY	LOWEST_MAX_SALARY
1	40000	5000

9. GROUP BY: total employees per department

```
SELECT DEPARTMENT_ID, COUNT(*) AS TOTAL_EMPLOYEES  
FROM EMPLOYEES GROUP BY DEPARTMENT_ID;
```

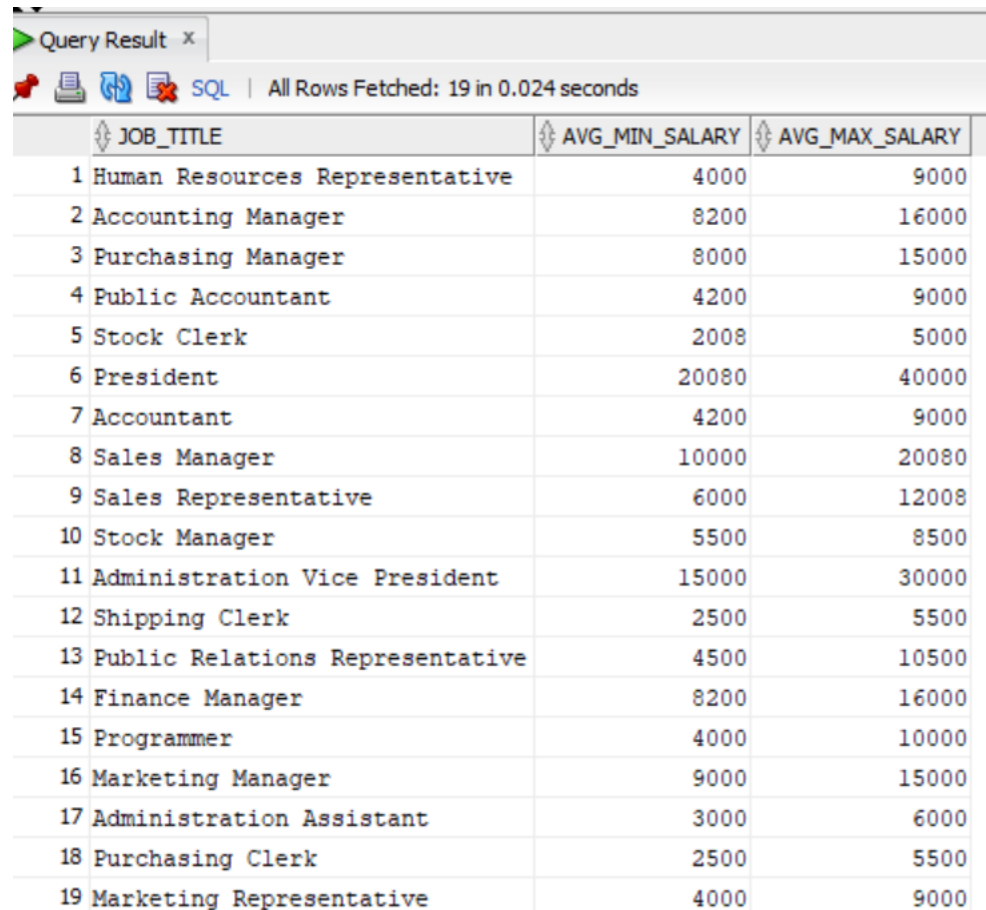


The screenshot shows a SQL query window with the following text: `SELECT DEPARTMENT_ID, COUNT(*) AS TOTAL_EMPLOYEES FROM EMPLOYEES GROUP BY DEPARTMENT_ID;`. Below the query, a 'Query Result' window displays the results. It shows 12 rows, each representing a department. The columns are 'DEPARTMENT_ID' and 'TOTAL_EMPLOYEES'. The status bar indicates 'All Rows Fetched: 12 in 0.012 seconds'.

DEPARTMENT_ID	TOTAL_EMPLOYEES
50	45
40	1
110	2
90	3
30	6
70	1
(null)	1
10	1
20	2
60	5
100	6
80	34

10. GROUP BY: average min and max salary per job

```
SELECT JOB_TITLE,  
  
       AVG(MIN_SALARY) AS AVG_MIN_SALARY,  
  
       AVG(MAX_SALARY) AS AVG_MAX_SALARY  
  
FROM JOBS GROUP BY JOB_TITLE;
```



Query Result x

SQL | All Rows Fetched: 19 in 0.024 seconds

JOB_TITLE	AVG_MIN_SALARY	AVG_MAX_SALARY
1 Human Resources Representative	4000	9000
2 Accounting Manager	8200	16000
3 Purchasing Manager	8000	15000
4 Public Accountant	4200	9000
5 Stock Clerk	2008	5000
6 President	20080	40000
7 Accountant	4200	9000
8 Sales Manager	10000	20080
9 Sales Representative	6000	12008
10 Stock Manager	5500	8500
11 Administration Vice President	15000	30000
12 Shipping Clerk	2500	5500
13 Public Relations Representative	4500	10500
14 Finance Manager	8200	16000
15 Programmer	4000	10000
16 Marketing Manager	9000	15000
17 Administration Assistant	3000	6000
18 Purchasing Clerk	2500	5500
19 Marketing Representative	4000	9000

11. HAVING: departments with more than 5 employees

```
SELECT DEPARTMENT_ID, COUNT(*) AS TOTAL_EMPLOYEES  
  
FROM EMPLOYEES GROUP BY DEPARTMENT_ID  
  
HAVING COUNT(*) > 5;
```



Query Result x

SQL | All Rows Fetched: 4 in 0.013 seconds

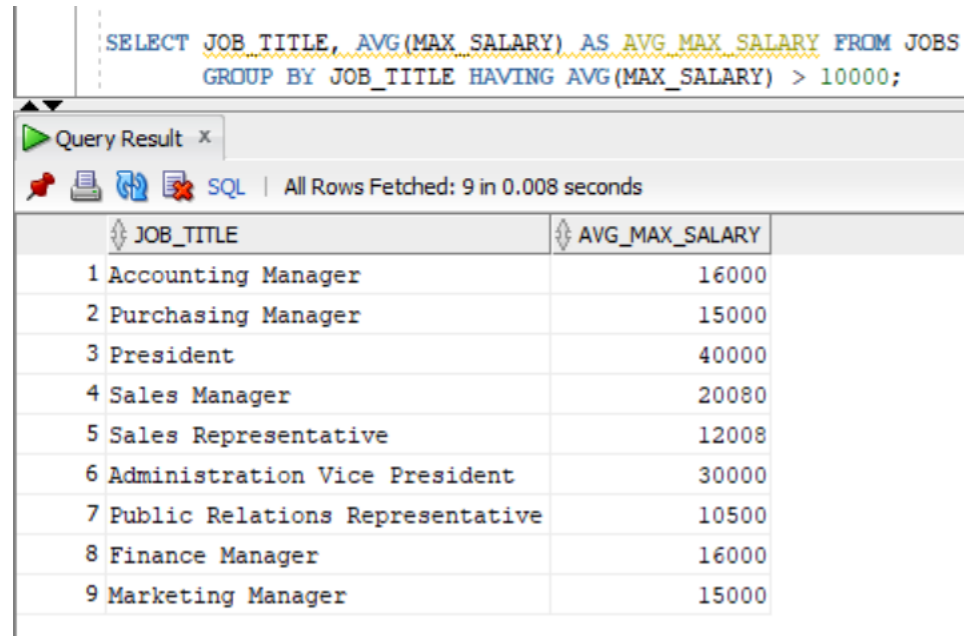
DEPARTMENT_ID	TOTAL_EMPLOYEES
1	50
2	30
3	100
4	80

12. HAVING: jobs where average max salary exceeds 10000

SELECT JOB_TITLE,

AVG(MAX_SALARY) AS AVG_MAX_SALARY

FROM JOBS GROUP BY JOB_TITLE HAVING AVG(MAX_SALARY) > 10000;



SELECT JOB_TITLE, AVG(MAX_SALARY) AS AVG_MAX_SALARY FROM JOBS
GROUP BY JOB_TITLE HAVING AVG(MAX_SALARY) > 10000;

Query Result x

SQL | All Rows Fetched: 9 in 0.008 seconds

JOB_TITLE	AVG_MAX_SALARY
1 Accounting Manager	16000
2 Purchasing Manager	15000
3 President	40000
4 Sales Manager	20080
5 Sales Representative	12008
6 Administration Vice President	30000
7 Public Relations Representative	10500
8 Finance Manager	16000
9 Marketing Manager	15000

ALL THE QUERIES:

```
SELECT COUNT(*) AS TOTAL_EMPLOYEES FROM EMPLOYEES;

SELECT COUNT(*) AS TOTAL_DEPARTMENTS FROM DEPARTMENTS;

SELECT SUM(SALARY) AS TOTAL_SALARY FROM EMPLOYEES;

SELECT SUM(COMMISSION_PCT) AS TOTAL_COMMISSION FROM EMPLOYEES;

SELECT AVG(SALARY) AS AVERAGE_SALARY FROM EMPLOYEES;

SELECT AVG(MIN_SALARY) AS AVG_MIN_SALARY, AVG(MAX_SALARY) AS
AVG_MAX_SALARY FROM JOBS;

SELECT MAX(SALARY) AS HIGHEST_SALARY, MIN(SALARY) AS LOWEST_SALARY
FROM EMPLOYEES;

SELECT MAX(MAX_SALARY) AS HIGHEST_MAX_SALARY,
      MIN(MAX_SALARY) AS LOWEST_MAX_SALARY FROM JOBS;

SELECT DEPARTMENT_ID, COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEES GROUP BY DEPARTMENT_ID;

SELECT JOB_TITLE,
      AVG(MIN_SALARY) AS AVG_MIN_SALARY,
      AVG(MAX_SALARY) AS AVG_MAX_SALARY
FROM JOBS GROUP BY JOB_TITLE;

SELECT DEPARTMENT_ID, COUNT(*) AS TOTAL_EMPLOYEES
FROM EMPLOYEES GROUP BY DEPARTMENT_ID
HAVING COUNT(*) > 5;

SELECT JOB_TITLE,
      AVG(MAX_SALARY) AS AVG_MAX_SALARY
FROM JOBS GROUP BY JOB_TITLE HAVING AVG(MAX_SALARY) > 10000;
```

